

Late Breaking Results: Fast System Technology Co-Optimization Framework for Emerging Technology Based on Graph Neural Networks

Tianliang Ma[†], Guangxi Fan[†], Xuguang Sun, Zhihui Deng, Kain Lu Low, Leilai Shao^{*}
Shanghai Jiao Tong University
Shanghai, China

ABSTRACT

This paper proposes a fast system technology co-optimization (STCO) framework that optimizes power, performance, and area (PPA) for next-generation IC design, addressing the challenges and opportunities presented by novel materials and device architectures. We focus on accelerating the technology level of STCO using AI techniques, by employing graph neural network (GNN)-based approaches for both TCAD simulation and cell library characterization, which are interconnected through a unified compact model, collectively achieving over a 100X speedup over traditional methods. These advancements enable comprehensive STCO iterations with runtime speedups ranging from 1.9X to 14.1X and supports both emerging and traditional technologies.

CCS CONCEPTS

• Hardware → Software tools for EDA.

ACM Reference Format:

Tianliang Ma[†], Guangxi Fan[†], Xuguang Sun, Zhihui Deng, Kain Lu Low, Leilai Shao. 2024. Late Breaking Results: Fast System Technology Co-Optimization Framework for Emerging Technology Based on Graph Neural Networks. In *61st ACM/IEEE Design Automation Conference (DAC '24)*, June 23–27, 2024, San Francisco, CA, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3649329.3663488>

1 INTRODUCTION

STCO integrates design and manufacturing early in chip development within the semiconductor industry, aiming to optimize performance, power efficiency, and yield through a unified consideration of design and manufacturing constraints. The iterative nature of traditional STCO approaches prolongs development, necessitating innovative solutions to expedite the process [1].

Our work advances the acceleration of technology level tasks within the STCO domain, the tasks at this level often encounter significant challenges, such as time-intensive TCAD simulations for device optimization [2], the lack of standardized models in transistor modeling [4], and the complexities in cell library characterization with emerging technologies and process variations [3]. Fig. 1 outlines our primary contributions to address these challenges, incorporating a TCAD surrogate model and cell characterization model based on GNN techniques. Additionally, parameter extraction is facilitated through our unified compact model, validated across technologies such as carbon nanotube (CNT), indium gallium zinc oxide (IGZO), and low-temperature polycrystalline silicon (LTPS). Furthermore, the efficacy of our framework is validated by

evaluating multiple designs using CNT technology, all conducted within the comprehensive STCO process depicted in Fig. 1.

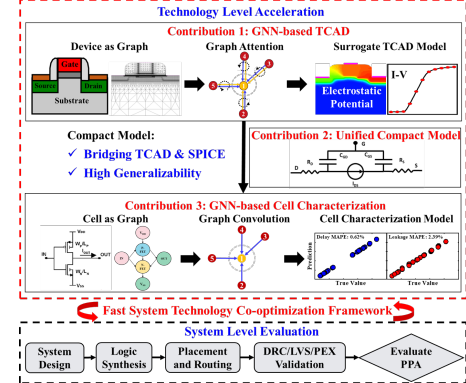


Figure 1: Fast STCO framework based on GNN.

2 FAST STCO FRAMEWORK

The framework accelerates the traditional process by employing a reinforcement learning (RL) agent to explore the design space across diverse benchmarks based on CNT technology. Our framework significantly enhances computational efficiency, reducing TCAD simulation time to 1.38 seconds and cell library characterization time to 8.88 seconds, beyond a shared environment setup time of 8.12 seconds for both processes. In contrast, commercial EDA tools require an average of 142.07 seconds for device simulations, as determined from a calibrated study involving 576 planar CNT devices with 2D TCAD simulations [5], and nearly 1900 seconds for cell library characterizations. These advancements demonstrate an acceleration exceeding 100 times for both individual tasks. For system evaluation, we utilized commercial tools for logic synthesis, placement & routing, and DRC & LVS checks. As illustrated in Table 1, the whole flow achieves a 1.9X to 14.1X acceleration over traditional flows per iteration. In addition, though initially tested on CNT technology, its adaptability allows easy application to other technologies like IGZO and LTPS.

Table 1: Runtime Comparison between two STCO flows

Benchmarks	Approximate Runtime(s)			Speedup(X)
	System Evaluation	Traditional STCO Framework	Ours	
s298	142	2184	160	13.6
s386	136	2178	154	14.1
s526	202	2244	220	10.2
s820	198	2240	216	10.4
s1196	223	2265	241	9.4
s1488	230	2272	248	9.2
16bit MAC	536	2578	554	4.7
32bit MAC	1270	3312	1288	2.6
Picorv32	939	2981	957	3.1
Darkrisvc	2250	4292	2268	1.9

Note: "Traditional STCO Framework" runtime combines system evaluation with commercial TCAD and cell library characterization times. "Ours" reflects system evaluation plus GNN-accelerated processes.

2.1 GNN-based Model for TCAD Simulation

The graph-based unified device encoding in Fig. 2 features embeddings at multiple levels. Material-level embedding comprises a one-hot vector for material type delineation and a parameter vector that details material properties and parameterizes physical models.

[†]Tianliang Ma and [†]Guangxi Fan contribute equally. ^{*}Corresponding author: Leilai Shao (leilaishao@sjtu.edu.cn). This work was supported by the National Key Research and Development Program of China: Design Technology Co-Optimization Methodology (grant number: 2023YFB4402700).

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

DAC '24, June 23–27, 2024, San Francisco, CA, USA

© 2024 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM ISBN 979-8-4007-0601-1/24/06

<https://doi.org/10.1145/3649329.3663488>

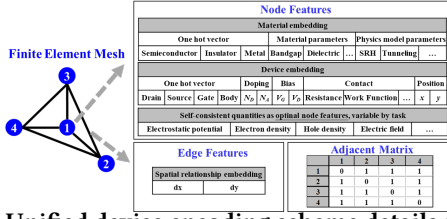


Figure 2: Unified device encoding scheme details node and edge features with task-specific self-consistent quantities.

Device-level embedding is designed to characterize individual devices, utilizing a one-hot vector for regional traits and an attribute vector to specify the operational parameters, encompassing doping, bias, and more. Spatial relationship embedding, serving as edge features, represents the relative positions between nodes. In addition, the encoding enables inclusion of task-specific self-consistent features such as charge density, tailored to the different tasks.

Utilizing node and graph regression, we established a Poisson emulator and an IV predictor for device characterization respectively. The Poisson emulator integrated charge density as an additional feature, whereas the node features of the IV predictor included both charge density and potential. Architecturally, the Poisson emulator employed a deep graph attention network with edge feature (RelGAT) and comprised approximately 1 million parameters, incorporating a 12-layer GAT with 2 attention heads and one multilayer perceptron (MLP). In contrast, the IV predictor utilized a shallower RelGAT model with about 0.15 million parameters, featuring a 3-layer, single-head GAT with a 4-layer MLP for prediction. Layer normalization was applied in the training of both models, enhancing model convergence and stability across the dataset of 50,000 independent devices. Table 2 showcases the effectiveness of the RelGAT model, demonstrating its high accuracy through validation and testing. To further assess the generalizability, an additional test on 32,000 unseen data samples was conducted, indicating a highly accurate surrogate TCAD model.

Table 2: MSE of Surrogate TCAD in the whole testing dataset

	Validation	Testing	Unseen(32K)	R^2 (32K)
Poisson Emulator	6.17×10^{-5}	7.02×10^{-5}	7.15×10^{-5}	0.9999
IV Predictor	1.67×10^{-3}	1.60×10^{-3}	1.78×10^{-3}	0.9999

2.2 Compact Model for Emerging Technologies

The compact model for emerging transistors is formulated to account for mobility variations in CNT, IGZO, and LTPS technologies due to charge drift in the presence of tail-distributed traps (TDTs) and variable range hopping (VRH) [4], as detailed in Eq. (1):

$$\mu = \begin{cases} \mu_0(V_G - V_{th})^\gamma, & \text{N-type TFT} \\ \mu_0(V_{th} - V_G)^\gamma, & \text{P-type TFT} \end{cases} \quad (1)$$

where V_{th} is the threshold voltage, γ is the field enhancement factor for mobility and μ_0 is defined as the effective mobility when $|V_G - V_{th}| = 1$. Validation against measured I-V curves from CNT, LTPS, and IGZO based devices is shown in Fig. 3.

2.3 GNN-based Cell Characterization Model

Our methodology employs a two-stage process. Initially, we developed a comprehensive cell library comprising 35 types of combinational and sequential cells and utilized transistor-level SPICE simulation to generate extensive cell datasets for training and testing, encompassing nine metrics including delay, output slew, leakage power, capacitance (maximum capacitance of each input pin), flip power (dynamic power generated when both input and output pins are flipped), non-flip power (dynamic power generated when only the input pins are flipped, while the state of the output pin remains unchanged), minimum setup/hold time, and pulse width (for sequential cells only) under various corners and technology

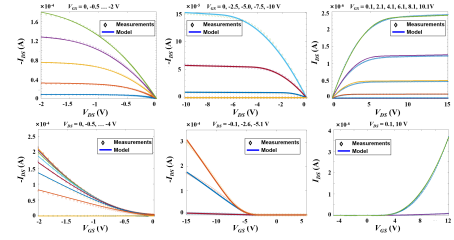


Figure 3: Validations of the proposed TFT model with measured I-V curves: (a) CNT-TFT with $L = 25\mu\text{m}$ and $W = 125\mu\text{m}$; (b) LTPS-TFT with $L = 16\mu\text{m}$ and $W = 40\mu\text{m}$; (c) IGZO-TFT with $L = 20\mu\text{m}$ and $W = 30\mu\text{m}$.

settings. Next, we adopted a 3-layer graph convolutional network (GCN) combined with an additional 2-layer MLP to establish our framework.

In our study of emerging technology, we utilized the unified compact model and specifically focused on analyzing the variation of supply voltage (V_{DD}), threshold voltage (V_{th}), and gate unit capacitance (C_{ox}). The definitions of node features are presented in Table 3. Here, "Current_state" and "Next_state" are two features related to the input pin state, with "1" denoting a high level and "0" representing a low level, respectively. "Input_slew" denotes the transition time of input signal, "Output_load" denotes capacitive load on cell output pins. For cell metrics that do not have relationship with some bits in the vector, these bits will be set to "0".

Table 3: Node feature definition for emerging technology

	IN	OUT	N-FET	P-FET	V_{DD}	V_{SS}
Bit0	0	0	0	0	0	0
Bit1	0	1	1	1	0	0
Bit2	1	0	1	1	0	1
Bit3	0	0	-1	1	0	0
Bit4	0	0	0	0	V_{DD}	0
Bit5	0	0	Width	Width	0	0
Bit6	0	0	Gate Unit Capacitance	Gate Unit Capacitance	0	0
Bit7	0	0	V_{th}	V_{th}	0	0
Bit8	Input_slew	0	0	0	0	0
Bit9	0	Output_load	0	0	0	0
Bit10	Current_state	0	0	0	0	0
Bit11	Next_state	0	0	0	0	0

As detailed in Table 4, valid data from 125 corners were for training while from 512 corners for testing, the model demonstrated high accuracy in experiments. The model exhibits a relatively large percentage error in predicting extremely low dynamic power consumption which needs to be improved in our future work.

Table 4: MAPEs of cell library prediction in testing dataset

	LTPS	CNT	Number of Data Points
Delay	0.47%	0.62%	696320
Output Slew	0.79%	0.83%	696320
Capacitance	0.18%	0.21%	70656
Flip Power	5.74%	4.96%	696320
Non-flip Power	3.36%	5.60%	393216
Leakage Power	2.78%	2.39%	165888
Minimum Pulse Width	1.20%	1.67%	8192
Minimum Setup	0.50%	0.27%	16384
Minimum Hold	0.45%	0.38%	16384

REFERENCES

- [1] El Mehdi Bazizi, Ashish Pal, Jongchol Kim, Liu Jiang, Vinod Reddy, Blessy Alexander, and Buvna Ayyagari-Sangamalli. 2021. Materials to systems co-optimization platform for rapid technology development targeting future generation CMOS nodes. *IEEE Transactions on Electron Devices* 68, 11 (2021), 5358–5363.
- [2] Wonik Jang, Sanghoon Myung, Jae Myung Choe, Young-Gu Kim, and Dae Sin Kim. 2023. Tcad device simulation with graph neural network. *IEEE Electron Device Letters* (2023).
- [3] Florian Klemme, Yogesh Chauhan, Jörg Henkel, and Hussam Amrouch. 2020. Cell library characterization using machine learning for design technology co-optimization. In *Proceedings of the 39th International Conference on Computer-Aided Design*. 1–9.
- [4] Leilai Shao, Ting Lei, Tsung-Ching Huang, Sicheng Li, Ta-Ya Chu, Man Wong, Raymond Beausoleil, Zhenan Bao, and Kwang-Ting Cheng. 2019. Compact modeling of thin-film transistors for flexible hybrid IoT design. *IEEE Design & Test* 36, 4 (2019), 6–14.
- [5] Lin Xu, Chenguang Qiu, Lian-mao Peng, and Zhiyong Zhang. 2021. Suppression of leakage current in carbon nanotube field-effect transistors. *Nano Research* 14 (2021), 976–981.